Lab 1: Introduction to MLOps & Reproducibility

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Task 1: Environment Isolation

Objective: Create an isolated Python environment, install specific versions of scikit-learn and pandas, and freeze the dependencies into `requirements.txt`.

Steps / commands

```
conda create -n mlops-lab python=3.11 -y
conda activate mlops-lab
pip install scikit-learn==1.5.2 pandas==2.1.4
pip list
pip freeze > requirements.txt
```

Result

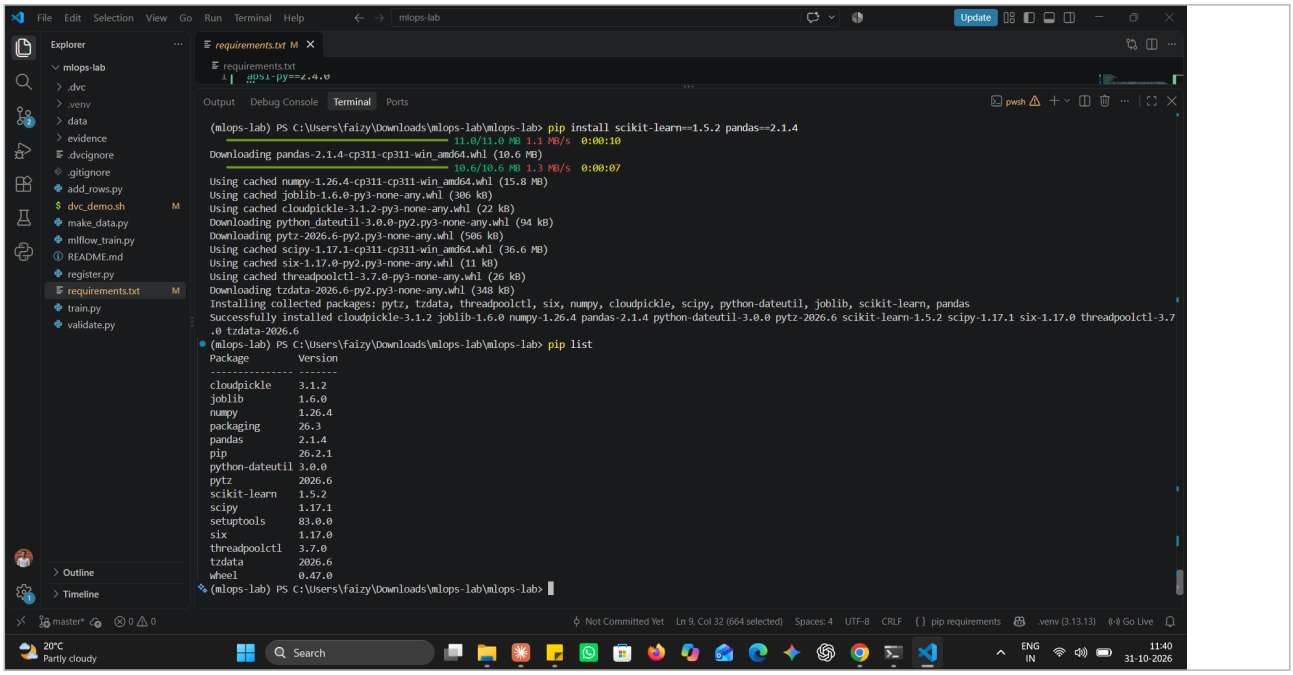


Figure 1: Installing scikit-learn 1.5.2 and pandas 2.1.4 in the new isolated environment *(mlops-lab)*, followed by *pip list* showing only a small set of packages.

Frozen dependencies (`requirements.txt`)

```
cloudpickle==3.1.2      joblib==1.6.0           numpy==1.26.4
packaging==26.3         pandas==2.1.4           python-dateutil==3.0.0
pytz==2026.6            scikit-learn==1.5.2     scipy==1.17.1
six==1.17.0             threadpoolctl==3.7.0    tzdata==2026.6
```

Explanation: A fresh environment was created so the project does not depend on the hundreds of packages in the base installation. Exact versions were pinned (`scikit-learn==1.5.2`, `pandas==2.1.4`) so that anyone can rebuild the same environment with `pip install -r requirements.txt`. pandas 2.1.4 was chosen because Great Expectations 1.2.0 (used in Lab 2) requires pandas below 2.2.

Task 2: Code Reproducibility

Objective: Train a basic decision tree on a sample dataset with fixed random seeds and assert that three separate executions give exactly the same accuracy.

Approach

• Dataset: Iris (built into scikit-learn).
• `random_state=42` is fixed in both `train_test_split` and `DecisionTreeClassifier`, the two places where randomness enters.
• The script launches itself three times as separate Python processes (`subprocess`) and collects the accuracy printed by each, to 10 decimal places.
• `assert len(set(accs)) == 1` fails if any run differs.

Script: train.py

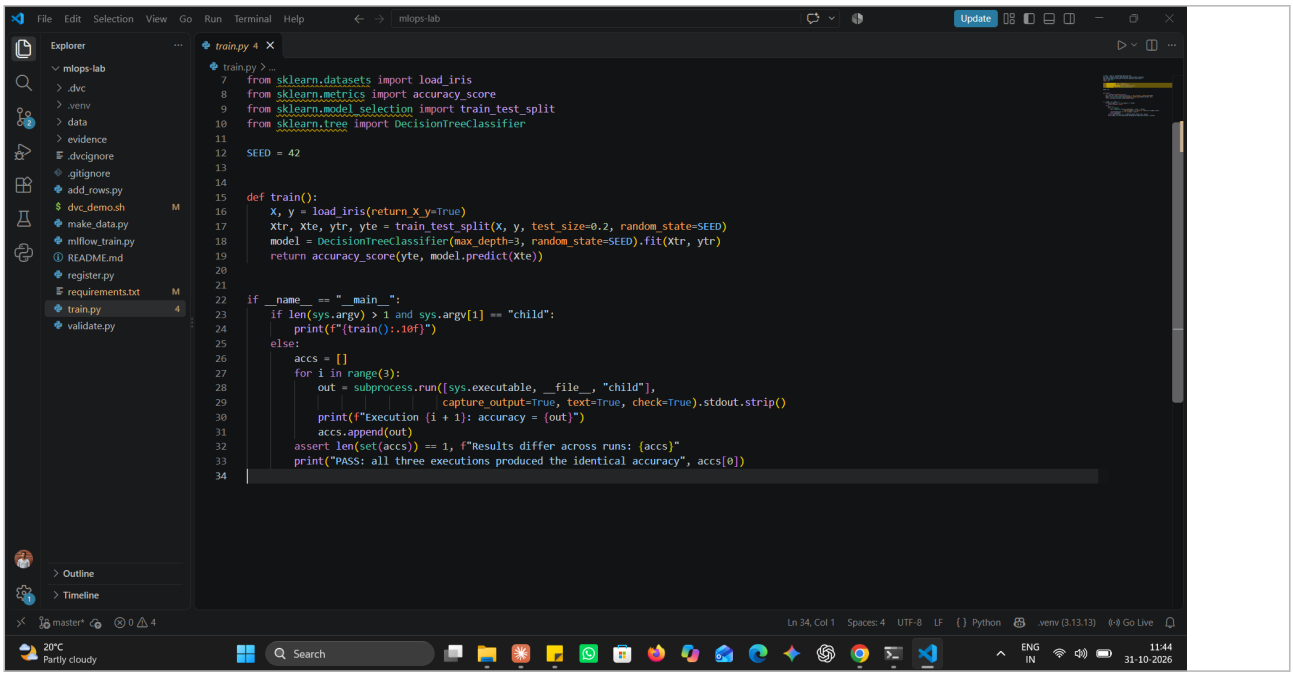


Figure 2: *train.py* in VS Code: fixed *SEED = 42*, the training function, and the three-run assertion.

Task 2 (continued): Execution output

Command

```
python train.py
```

Result

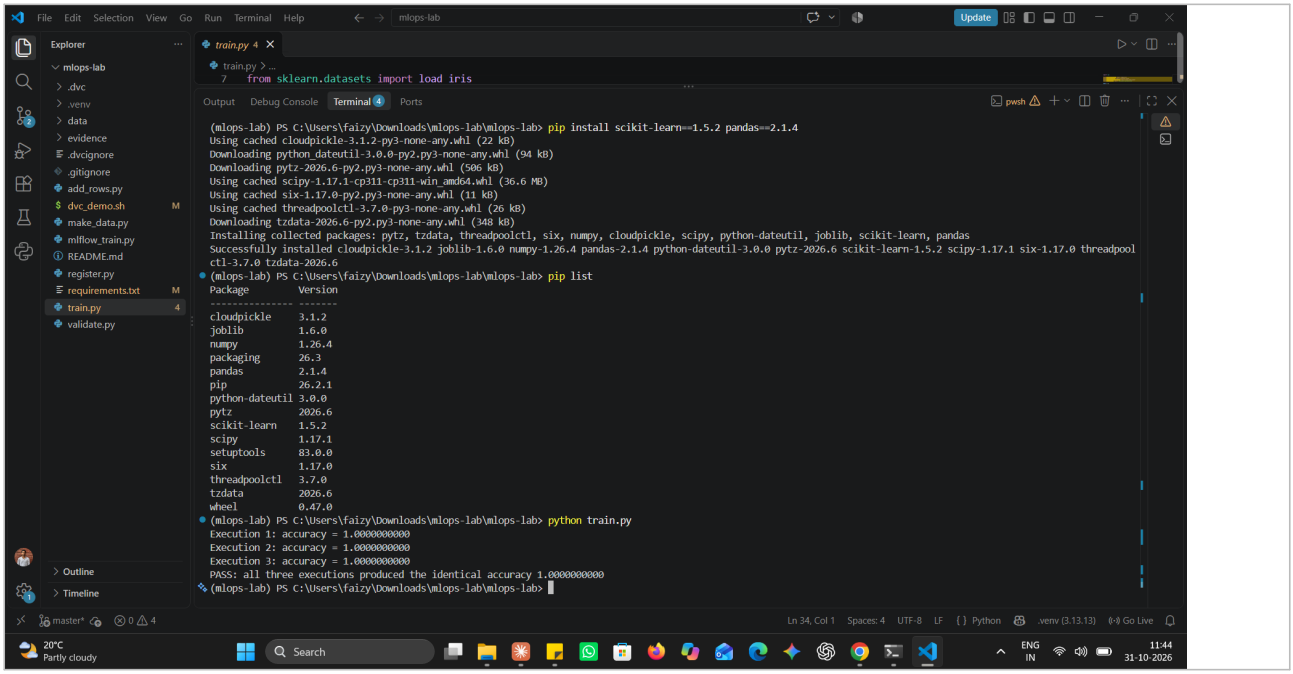


Figure 3: Output of *python train.py*: all three executions give accuracy = 1.0000000000 and the assertion passes.

Output

```
Execution 1: accuracy = 1.0000000000
Execution 2: accuracy = 1.0000000000
Execution 3: accuracy = 1.0000000000
PASS: all three executions produced the identical accuracy 1.0000000000
```

Conclusion: Because the random seeds are fixed, the train/test split and the tree are identical in every run, so the accuracy is exactly the same across three independent executions. The assertion passed, confirming the experiment is reproducible.